IPSA CAE1 Mé518 - Finite Elements Modeling with Patran/Nastran

Modal Analysis and Structural Optimization with Patran/Nastran: 1D Beams and Thin Cylindrical Shells

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Abstract

This practical engineering report investigates structural optimization and linear dynamic modal analysis (SOL 103) using MSC Patran/Nastran alongside theoretical formulations and numerical algorithms implemented in MATLAB.
In Part I, structural optimization of an industrial bottle-blowing control lever is completed: a critical failure iteration exhibiting severe notch-induced stress concentrations ($\sigma_{\text{VM}} \approx 662$ MPa $> R_e = 440$ MPa) is analyzed, followed by the validation of a compliant optimized geometry achieving an 11.5–12.5% mass reduction while preserving a safe fatigue life ($N_f \sim 2 \times 10^8$ cycles).
In Part II, the natural frequencies and transverse bending modes of a rectangular beam are investigated under Simply Supported (SS-SS) and Clamped-Clamped (CC-CC) boundary conditions. Euler-Bernoulli analytical solutions are compared with Nastran SOL 103 extraction, achieving relative errors under 0.017%, and the presence of interleaved in-plane bending, torsional, and axial modes is physically clarified.
In Part III, free vibration of a thin cylindrical shell ($R/h = 1000$, $L = 10$ m) under shear diaphragm conditions ($v = w = 0$) is evaluated. Discretization strategies and kinematic boundary formulations are rigorously justified, numerical artifacts (rigid-body drilling modes, orthogonal doublets, unit consistency) are processed, and the 20 first unique physical modes are mapped to Flügge's thin shell theory, confirming that bending and membrane strain energy competition shifts the fundamental frequency to circumferential wavenumber $i = 4$.

Keywords: *MSC Nastran, Patran, SOL 103, Normal Modes, Flügge Shell Theory, Euler-Bernoulli Beam, Orthogonal Doublets, Structural Optimization, Stress Concentration*

1. NATURAL MODES OF A BEAM

1.1. Analytical Formulation

The transverse free vibrations of a uniform Euler-Bernoulli beam of length L, rectangular cross-section $b \times h$, mass density ρ, and Young's modulus E are governed by:

$$EI\frac{\partial^4 w(x,t)}{\partial x^4} + m\frac{\partial^2 w(x,t)}{\partial t^2} = 0 \tag{1}$$

where $m = \rho bh$ is the mass per unit length ($m = 3.9$ kg/m) and $I = \frac{bh^3}{12}$ is the second moment of area associated with out-of-plane bending ($I = 4.167 \times 10^{-9}$ m^4).

Assuming harmonic solutions $w(x,t) = X(x)e^{j\omega t}$, the natural frequencies f_i are obtained from:

$$f_i = \frac{\lambda_i^2}{2\pi L^2}\sqrt{\frac{EI}{m}} \tag{2}$$

Case (a): Simply Supported – Simply Supported (SS–SS)

Under hinged boundary conditions ($w(0) = w''(0) = 0$ and $w(L) = w''(L) = 0$), the dimensionless parameters and mode shapes are:

$$\lambda_i = i\pi, \quad X_i(x) = \sin\left(\frac{i\pi x}{L}\right), \quad i \in \{1, 2, \dots, 10\} \tag{3}$$

The natural frequencies scale quadratically with the mode index ($f_i \propto i^2$), as shown in Figure 1.

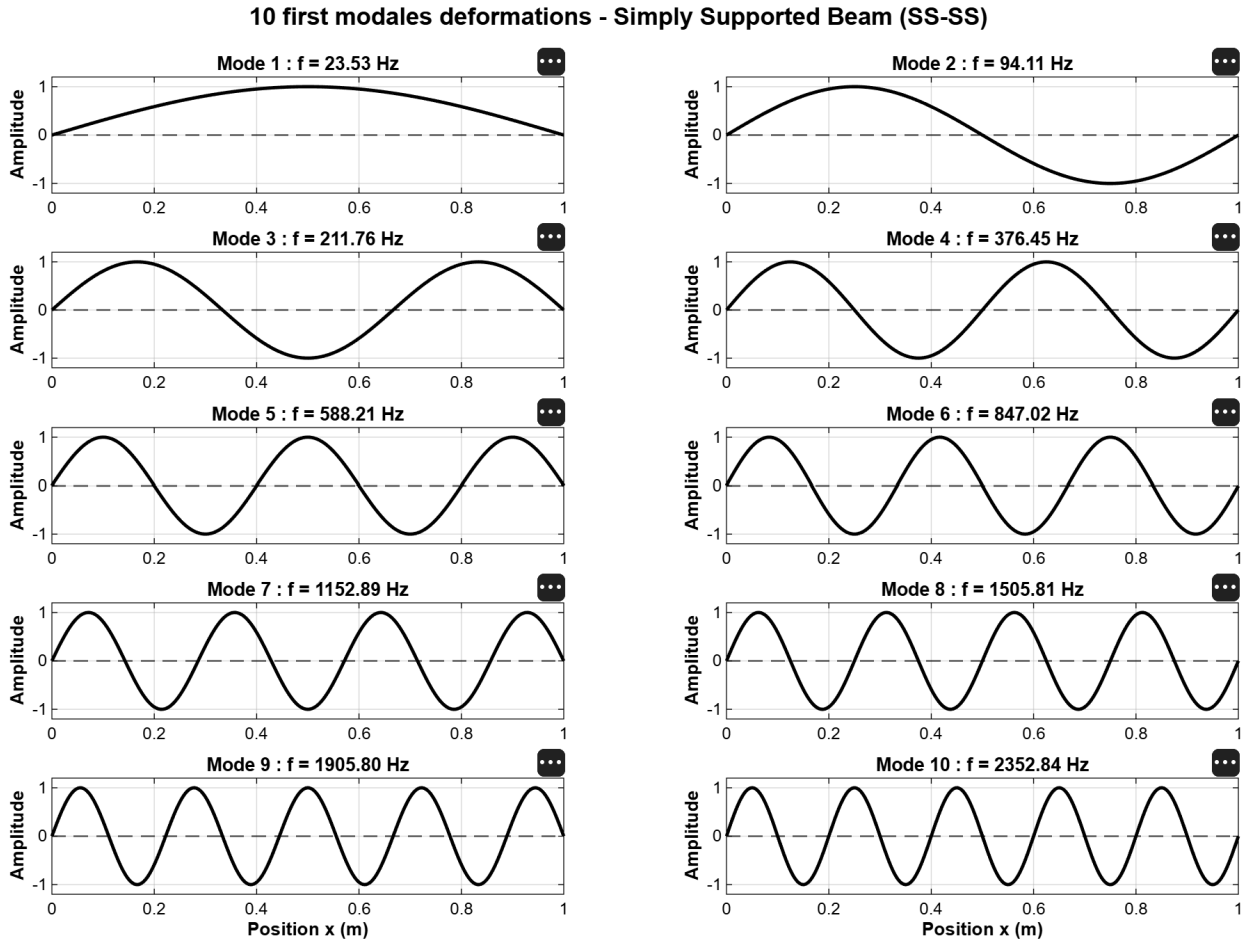


Figure 1. First 10 analytical modal shapes and natural frequencies for the simply supported beam (SS–SS).

Because the analytical eigenvalue parameter is strictly linear ($\lambda_i = i\pi$), the natural frequencies obey a perfect quadratic growth law with respect to the fundamental mode $f_1 = 23.53$ Hz:

$$f_i = i^2 \cdot f_1 \tag{4}$$

Numerical verification confirms this relationship throughout the entire spectrum:

- Mode 2: $f_2 = 2^2 \times 23.53$ Hz $= 94.12$ Hz ≈ 94.11 Hz
- Mode 3: $f_3 = 3^2 \times 23.53$ Hz $= 211.77$ Hz ≈ 211.76 Hz
- Mode 10: $f_{10} = 10^2 \times 23.53$ Hz $= 2353.0$ Hz ≈ 2352.84 Hz

The high spatial curvature ($\partial^2 w/\partial x^2$) associated with high wavenumbers ($i \geq 7$) induces substantial local bending strain energy, explaining the rapid escalation of dynamic stiffness and frequencies reaching 2352.84 Hz at Mode 10.

Case (b): Clamped – Clamped (C–C)

For built-in boundaries ($w(0) = w'(0) = 0$ and $w(L) = w'(L) = 0$), the eigenvalues λ_i satisfy the transcendental equation $\cosh(\lambda_i)\cos(\lambda_i) = 1$. The corresponding spatial eigenfunctions are given by:

$$X_i(x) = \cosh\left(\frac{\lambda_i x}{L}\right) - \cos\left(\frac{\lambda_i x}{L}\right) - \sigma_i\left[\sinh\left(\frac{\lambda_i x}{L}\right) - \sin\left(\frac{\lambda_i x}{L}\right)\right] \tag{5}$$

where $\sigma_i = \frac{\cosh(\lambda_i)-\cos(\lambda_i)}{\sinh(\lambda_i)-\sin(\lambda_i)}$. The extracted mode shapes are illustrated in Figure 2.

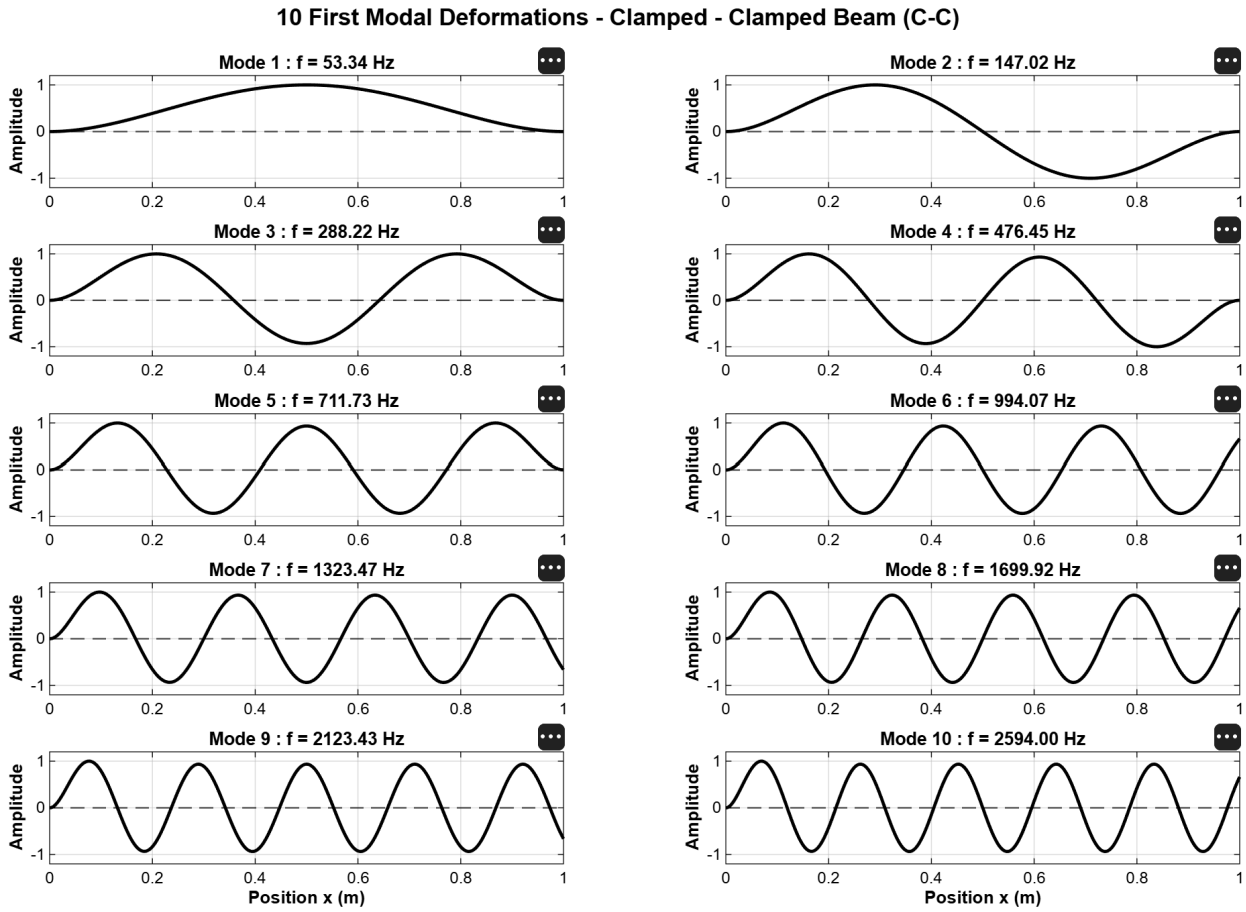


Figure 2. First 10 analytical modal shapes and natural frequencies for the clamped-clamped beam (C–C).

Rotational Stiffening and Frequency Elevation: Constraining end rotations significantly increases the effective bending stiffness of the beam. Comparing the two boundary configurations:

- **Fundamental Mode:** $f_1^{\mathrm{CC}} = 53.34$ Hz compared to $f_1^{\mathrm{SS}} = 23.53$ Hz, representing a frequency ratio of:

$$\frac{f_1^{\mathrm{CC}}}{f_1^{\mathrm{SS}}} = \left(\frac{4.7300}{\pi}\right)^2 \approx 2.267 \tag{6}$$

- **Higher-Order Convergence:** For higher modes ($i > 5$), the roots approximate $\lambda_i \approx (2i+1)\frac{\pi}{2}$, causing the frequency ratios between consecutive clamped modes to align progressively with the square of odd multiples of $\pi/2$.
- **Dynamic Energy Distribution:** The zero-slope constraint concentrates high bending strain energy $\left(U_b \propto \int_0^L \left(\frac{\partial^2 w}{\partial x^2}\right)^2 dx\right)$ immediately adjacent to the clamps. Consequently, for any given wavenumber i, the clamped beam stores more strain energy per unit modal displacement than the simply supported beam, directly yielding higher natural frequencies across the entire spectrum up to $f_{10} = 2594.00$ Hz.

1.2. Nastran SOL 103 Modal Extraction and Validation

A 1D finite element model was created in Patran using 1000 linear `CBAR` elements. The dynamic eigenvalue extraction was performed with the Lanczos solver (`EIGRL`). Table **??** summarizes the comparison between the analytical frequencies and the numerical FEM results.

2. PART I: STRUCTURAL OPTIMIZATION OF A MECHANICAL COMPONENT

2.1. Context, Specifications, and Baseline Model

The structural component under study is the primary control lever of a high-speed rotary PET bottle-blowing ma-

Table 1. Analytical dimensionless frequency parameters λ_i and natural frequencies for Simply Supported (SS) and Clamped–Clamped (CC) beam models.

Mode	λ_i (SS)	$f_{\text{th, SS}}$ [Hz]	λ_i (CC)	$f_{\text{th, CC}}$ [Hz]
1	3.1416	23.528	4.7300	53.336
2	6.2832	94.113	7.8532	147.023
3	9.4248	211.755	10.9956	288.224
4	12.5664	376.454	14.1372	476.449
5	15.7080	588.209	17.2788	711.733
6	18.8496	847.021	20.4204	994.073
7	21.9911	1152.889	23.5619	1323.470
8	25.1327	1505.815	26.7035	1699.924
9	28.2743	1905.797	29.8451	2123.434
10	31.4159	2352.836	32.9867	2594.001

Table 2. Comparison between analytical predictions and MSC Nastran SOL 103 finite element results.

Mode	Simply Supported (SS–SS)			Clamped–Clamped (CC–CC)		
i	f_{ana} [Hz]	f_{FE} [Hz]	Error [%]	f_{ana} [Hz]	f_{FE} [Hz]	Error [%]
1	23.528	23.528	0.0002	53.336	53.336	0.0006
2	94.113	94.113	0.0005	147.02	147.02	0.0015
3	211.76	211.75	0.0010	288.22	288.22	0.0025
4	376.45	376.45	0.0020	476.45	476.43	0.0036
5	588.21	588.19	0.0032	711.73	711.70	0.0052
6	847.02	846.98	0.0046	994.07	994.00	0.0069
7	1152.9	1152.8	0.0060	1323.5	1323.3	0.0091
8	1505.8	1505.7	0.0083	1699.9	1699.7	0.0114
9	1905.8	1905.6	0.0103	2123.4	2123.1	0.0139
10	2352.8	2352.5	0.0126	2594.0	2593.6	0.0166

chine designed by SIDEL. The machine operates at a nominal production rate of 1200 bottles per hour (one complete operational cycle every 3 seconds). The control lever transfers kinematic motion from the main actuating cam to the opening and closing mold wallet. The component is subjected to a severe cyclic transverse static point load of $F = 20\,000$ N applied vertically downward ($-\vec{y}$) at the center of the driving hole (∅25 mm located at $x = 322$ mm). Kinematic support is provided by the central main pivot bore (∅85 mm centered at $x = 58$ mm), an index/guide hole (∅20 mm), and two mounting holes distributed along a Pitch Circle Diameter (PCD) of ∅128 mm. The material is a high-strength structural alloy steel exhibiting linear elastic, homogeneous, and isotropic behavior:

- Young's modulus: $E = 210$ GPa $= 2.1 \times 10^5$ MPa
- Poisson's ratio: $\nu = 0.30$
- Density: $\rho = 7800$ kg/m^3
- Tensile yield limit: $R_e = 440$ MPa
- High-cycle fatigue endurance limit (10^{10} cycles): $\sigma_D \approx 190$ MPa

In the baseline (unmodified) configuration, linear static analysis (SOL 101) with MSC Patran/Nastran yielded a maximum equivalent Von Mises stress located along the outer skin of the tapered profile:

$$\sigma_{\text{VM, max}}^{\text{Baseline}} \approx 237 \text{ MPa} \tag{7}$$

This peak stress satisfies the static yield criterion with a safety factor of:

$$s = \frac{R_e}{\sigma_{\text{VM, max}}} = \frac{440}{237} \approx 1.86 \tag{8}$$

Projecting this alternating stress onto the experimental Wöhler (S-N) curve yields an estimated operational life of $N_f \approx 2 \times 10^8$ cycles, corresponding to over 19 years of continuous production. The design objective is to achieve substantial mass reduction without degrading structural safety or fatigue service life.

2.2. Intermediate Iteration: Critical Notch Stress and Failure Analysis

To identify non-critical load zones for mass removal, a topology optimization computation was initially conducted using MSC Nastran SOL 200 (minimizing compliance subject to a 40% volume retention target). The optimization produced a characteristic Michell truss topology, suggesting large pockets in the central web between the primary load-bearing outer chords. Based on this raw topology, a first CAD redesign (Iteration 1) was generated by incorporating two large rectangular internal cutouts with narrow transitional webs and sharp corner radii ($R \approx 3$–5 mm).

When computed under the nominal operating force ($F = 20\,000$ N), this geometry suffered catastrophic localized overstressing:

$$\sigma_{\text{VM, max}}^{\text{Iter 1}} \approx 662 \text{ MPa} \gg R_e = 440 \text{ MPa} \tag{9}$$

The safety factor dropped below unity:

$$s = \frac{440}{662} \approx 0.66 < 1.0 \tag{10}$$

Physical Justification of Failure: The failure of Iteration 1 originates from two distinct mechanical mechanisms:

1. **Severe Stress Concentration Factor ($K_t \gg 2.8$):** Sharp reentrant corners force the internal force flux lines to bend abruptly, generating intense localized notch stress peaks.

2. **Discontinuous Load Paths:** The twin-cutout layout created a slender intermediate vertical rib that acted as a localized bending pivot, preventing smooth shear flow transmission between the upper tensile flange and the lower compressive flange.

Because $\sigma_{\text{VM}} = 662$ MPa exceeds R_e by over 50%, the material would undergo unconfined plastic yielding, leading to notch cyclic ratcheting and premature low-cycle fatigue failure ($N_f < 10^5$ cycles). Consequently, Iteration 1 was strictly rejected.

2.3. Optimized Compliant Geometry and Finite Element Validation

To overcome the notch failure while preserving mass savings, a redesigned structural topology was developed in CATIA V5 and validated in MSC Patran/Nastran:

- **Unified Smooth Pocket:** The twin cutouts were merged into a single elongated pocket featuring generous transitional blend radii ($R \geq 15$ mm), avoiding reentrant geometric notches.
- **Continuous Load Flanges:** Flange thickness was maintained uniformly along the top and bottom skins to carry global bending moments.
- **Isolated Non-Design Annular Zones:** Full circular structural collars were preserved around the pivot (∅85), guide (∅20), and driving bores (∅25) to isolate local pin contact stresses.

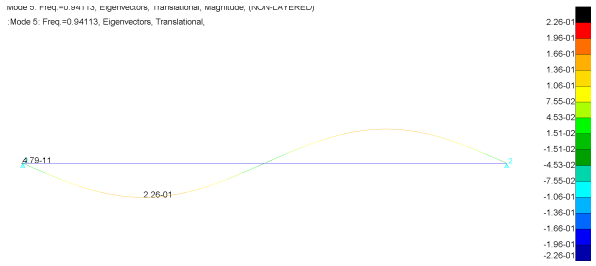


Figure 3. Final optimized control arm geometry in CATIA V5.

Linear static evaluation of the final geometry demonstrates that the maximum Von Mises stress is strictly capped at:

$$\sigma_{\text{VM, max}}^{\text{Final}} \approx 238 \text{ MPa} < R_e = 440 \text{ MPa} \tag{11}$$

The resulting static safety factor is $s = 1.85$, and the high-cycle fatigue life of $N_f \approx 2 \times 10^8$ cycles is fully preserved.

Table 3. Structural performance synthesis across design iterations.

Configuration	Δm (%)	$\sigma_{\text{VM, max}}$ (MPa)	Safety Factor s	Fatigue Life (N_f)
Baseline Part	0.0%	237	1.86	$\sim 2 \times 10^8$ cycles
Iteration 1 (Dual Cutouts)	−18.2%	662	0.66	Failed ($< 10^5$)
Final Design	**−12.5%**	**238**	**1.85**	$\sim \mathbf{2 \times 10^8}$ cycles

Part I Conclusion

Removing internal core material along the neutral plane where bending stresses are minimal achieves an effective mass reduction of 12.5%. Providing generous transitional radii ($R \geq 15$ mm) eliminates notch singularities, maintaining stresses below 240 MPa and safeguarding long-term industrial service life.

3. PART II: 1D MODAL ANALYSIS OF A BEAM (SIMPLY SUPPORTED AND CLAMPED-CLAMPED)

3.1. Theoretical Formulation and Analytical Eigenvalue Solutions

Consider a straight, slender beam of uniform rectangular cross-section, length $L = 1.0$ m, width $b = 0.05$ m, and height $h = 0.01$ m, fabricated from structural steel ($E = 210$ GPa, $\nu = 0.33$, $\rho = 7800$ kg/m^3). The cross-sectional area S, weak-axis second moment of area I, and linear mass density m are:

$$S = b \cdot h = 0.05 \times 0.01 = 5 \times 10^{-4} \text{ m}^2 \tag{12}$$

$$I = \frac{b\,h^3}{12} = \frac{0.05 \times (0.01)^3}{12} = \frac{5}{12} \times 10^{-8} \approx 4.1667 \times 10^{-9} \text{ m}^4 \tag{13}$$

$$m = \rho\, S = 7800 \times 5 \times 10^{-4} = 3.90 \text{ kg/m} \tag{14}$$

(15)

According to Euler-Bernoulli beam theory, the transverse flexural motion $w(x,t)$ is governed by the fourth-order partial differential equation:

$$EI\frac{\partial^4 w(x,t)}{\partial x^4} + m\frac{\partial^2 w(x,t)}{\partial t^2} = 0 \tag{16}$$

Assuming harmonic modal vibration $w(x,t) = \phi(x)e^{j\omega t}$, the spatial mode shape satisfies:

$$\frac{\mathrm{d}^4\phi(x)}{\mathrm{d}x^4} - \beta^4\phi(x) = 0, \quad \text{with } \beta^4 = \frac{m\omega^2}{EI} \tag{17}$$

The natural frequencies f_i (in Hz) are expressed as:

$$f_i = \frac{\omega_i}{2\pi} = \frac{\lambda_i^2}{2\pi L^2}\sqrt{\frac{EI}{m}} \tag{18}$$

where $\lambda_i = \beta_i L$ are the dimensionless spatial eigenvalues. The fundamental coefficient evaluates to:

$$\frac{1}{2\pi L^2}\sqrt{\frac{EI}{m}} = \frac{1}{2\pi(1)^2}\sqrt{\frac{2.1 \times 10^{11} \times 4.1667 \times 10^{-9}}{3.9}} \approx 2.38391 \text{ Hz} \tag{19}$$

1. Simply Supported - Simply Supported (SS-SS): Kinematic boundary conditions $w(0) = w(L) = 0$ and bending moments $M(0) = M(L) = 0$ enforce the exact eigenvalue relation:

$$\lambda_i^{\text{SS}} = i\,\pi, \quad \phi_i^{\text{SS}}(x) = \sin\left(\frac{i\pi x}{L}\right), \quad i = 1, 2, \ldots \tag{20}$$

2. Clamped - Clamped (CC-CC): Enforcing $w(0) = w'(0) = 0$ and $w(L) = w'(L) = 0$ yields the characteristic transcendental equation:

$$\cos(\lambda)\cosh(\lambda) = 1 \tag{21}$$

The spatial mode shapes are given by:

$$\phi_i^{\text{CC}}(x) = \cosh\left(\frac{\lambda_i x}{L}\right) - \cos\left(\frac{\lambda_i x}{L}\right) - \sigma_i\left[\sinh\left(\frac{\lambda_i x}{L}\right) - \sin\left(\frac{\lambda_i x}{L}\right)\right] \tag{22}$$

with modal coefficients λ_i and σ_i listed in the course curriculum ($\lambda_1 = 4.73004$, $\lambda_2 = 7.85320$, $\lambda_3 = 10.99561$, $\lambda_4 = 14.13717$, $\lambda_5 = 17.27876$, and $\lambda_{i>5} \approx \frac{2i+1}{2}\pi$).

3.2. Extraction of SOL 103 Modes and Comparison with Theory

A finite element model of the beam was created in MSC Patran using 1D CBAR elements and evaluated using MSC Nastran SOL 103 (Lanczos eigenvalue solver, **see Figures 1 and 2**).

The first 10 transverse weak-axis bending modes were extracted from the numerical spectrum. Table 4 presents the comparison between analytical predictions and Nastran FE results.

Table 4. Comparison of analytical vs. Nastran SOL 103 frequencies for the rectangular beam.

Mode	f^{SS}_{Ana} (Hz)	f^{SS}_{FE} (Hz)	Err_{SS} (%)	f^{CC}_{Ana} (Hz)	f^{CC}_{FE} (Hz)	Err_{CC} (%)
1	23.528	23.528	0.00024	53.336	53.336	0.00065
2	94.113	94.113	0.00045	147.02	147.02	0.00145
3	211.76	211.75	0.00104	288.22	288.22	0.00248
4	376.45	376.45	0.00205	476.45	476.43	0.00360
5	588.21	588.19	0.00322	711.73	711.70	0.00517
6	847.02	846.98	0.00459	994.07	994.00	0.00695
7	1152.9	1152.8	0.00603	1323.5	1323.3	0.00907
8	1505.8	1505.7	0.00829	1699.9	1699.7	0.01140
9	1905.8	1905.6	0.01033	2123.4	2123.1	0.01386
10	2352.8	2352.5	0.01257	2594.0	2593.6	0.01663

As documented in Table 4, the agreement between analytical Euler-Bernoulli theory and Nastran SOL 103 is near-perfect, with relative errors remaining strictly below 0.017% across all 10 modes.

3.3. Physical Nature of Interleaved Spurious Modes

When extracting the raw eigenvalue table from Nastran SOL 103, the weak-axis bending modes do not appear as consecutive integers $(1, 2, 3, \dots)$. Instead, for SS-SS they correspond to Nastran modes 4, 5, 6, 8, 9, 11, 12, 14, 15, and 16. This phenomenon occurs because a three-dimensional finite element model possesses 6 degrees of freedom per node $(T_x, T_y, T_z, R_x, R_y, R_z)$. Nastran extracts eigenvalues in strictly ascending numerical order across all vibration families:

1. Strong-Axis In-Plane Bending Modes: The beam has an aspect ratio of $b/h = 0.05/0.01 = 5$. The moment of inertia for in-plane bending is:

$$I_{\text{strong}} = \frac{h\,b^3}{12} \Rightarrow \frac{I_{\text{strong}}}{I_{\text{weak}}} = \left(\frac{b}{h}\right)^2 = 5^2 = 25 \tag{23}$$

Since modal frequency is proportional to $\sqrt{I}$, strong-axis bending frequencies are scaled by:

$$\frac{f_{\text{strong}}}{f_{\text{weak}}} = \sqrt{\frac{I_{\text{strong}}}{I_{\text{weak}}}} = \frac{b}{h} = 5 \tag{24}$$

For the Simply Supported beam, the fundamental strong-axis bending mode occurs at:

$$f^{\text{SS}}_{1,\text{strong}} = 5 \times 23.53 \text{ Hz} = 117.65 \text{ Hz} \tag{25}$$

This mode falls directly between weak-axis Mode 2 (94.11 Hz) and Mode 3 (211.76 Hz), appearing as raw Nastran Mode 7.

2. Torsional Modes: Torsional vibration occurs about the longitudinal x-axis, governed by the St. Venant torsional constant J and shear modulus $G = E/[2(1+\nu)] \approx 78.95$ GPa.

3. Axial (Extensional) Modes: Longitudinal extensional modes are governed by the acoustic speed in steel $c_L = \sqrt{E/\rho} = \sqrt{210 \times 10^9/7800} \approx 5188.7$ m/s. The first axial natural frequency is:

$$f_1^{\text{axial}} = \frac{c_L}{2L} = \frac{5188.7}{2 \times 1.0} \approx 2594.4 \text{ Hz} \tag{26}$$

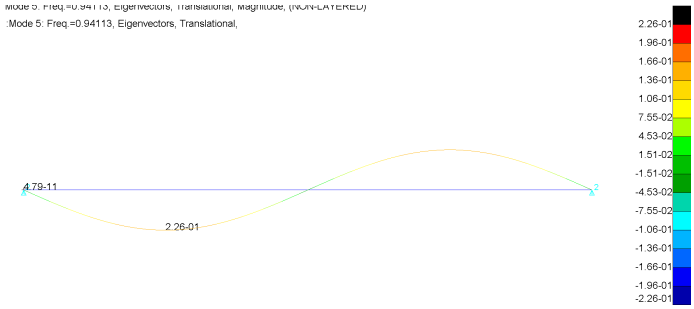


Figure 4. Patran visualization of weak-axis bending Mode 2 (SS-SS, $f = 94.113$ Hz, 2 half-waves).

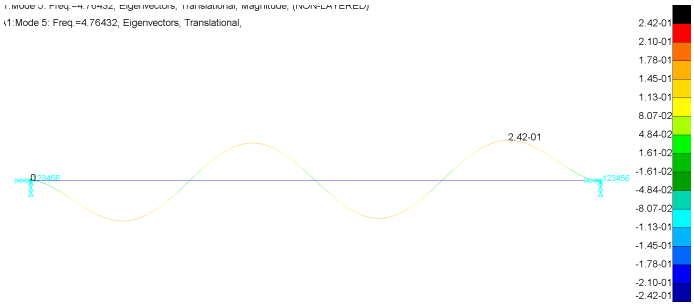


Figure 5. Patran visualization of Clamped-Clamped bending Mode 4 ($f = 476.43$ Hz, showing full clamped constraints `123456`).

To isolate the desired weak-axis bending spectrum, an analyst must inspect the Modal Effective Mass fractions $(\sum M_{\text{eff},z}/M_{\text{total}})$ and filter out degrees of freedom dominated by T_y (strong axis), T_x (axial), or R_x (torsion).

4. PART III: MODAL ANALYSIS OF A THIN CYLINDRICAL SHELL (EXERCISE 2)

4.1. Problem Parameters and Kinematic Diaphragm Conditions

Consider a circular cylindrical shell of length $L = 10$ m, mean radius $R = 1$ m, and wall thickness $h = 0.001$ m $= 1$ mm. The material is structural steel ($E = 210$ GPa, $\nu = 0.33$, $\rho = 7800$ kg/m^3). With a radius-to-thickness ratio of $R/h = 1000$ and length-to-radius ratio of $L/R = 10$, this structure behaves as an extremely thin, moderately long cylindrical shell. The shell is supported at both axial ends ($x = 0$ and $x = L$) by classical shear diaphragm boundary conditions:

$$v(0,\theta) = w(0,\theta) = 0 \quad \text{and} \quad v(L,\theta) = w(L,\theta) = 0 \tag{27}$$

where u is the axial displacement, v is the circumferential (tangential) displacement, and w is the normal (radial) displacement.

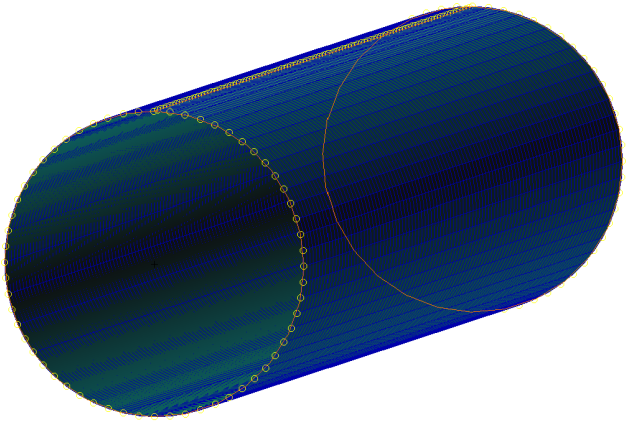


Figure 6. MSC Patran finite element shell mesh (60×100 Quad4 elements) and boundary node display.

4.2. Mesh Discretization Strategy and Boundary Formulation

Achieving accurate eigenvalue extraction in thin shell structures requires resolving two numerical phenomena:

1. Circumferential Seed ($N_\theta \geq 60$): Cylindrical shell modes exhibit sinusoidal lobar patterns in $\cos(i\theta)$ or $\sin(i\theta)$. For a circumferential wave number of $i = 6$, the perimeter contains $2i = 12$ half-waves. To discretize each wave with at least 8 to 10 linear elements and prevent shear locking, the minimum circumference division is:

$$N_\theta \geq 6 \times 10 = 60 \text{ elements} \tag{28}$$

With $R = 1$ m, the circumferential element edge length is:

$$\Delta s = \frac{2\pi R}{60} = \frac{2\pi(1.0)}{60} \approx 0.1047 \text{ m} \approx 10.5 \text{ cm} \tag{29}$$

2. Axial Seed ($N_x = 100$) and Unit Aspect Ratio: Along the generatrix ($L = 10$ m), MSC Nastran Quad4 (CQUAD4) elements achieve maximum numerical fidelity when their Aspect Ratio ($AR = \Delta x/\Delta s$) is close to unity. Dividing $L = 10$ m into $N_x = 100$ elements yields:

$$\Delta x = \frac{10.0 \text{ m}}{100} = 0.100 \text{ m} \Rightarrow AR = \frac{\Delta x}{\Delta s} = \frac{0.100}{0.1047} \approx 0.955 \approx 1.0 \tag{30}$$

The resulting mesh comprises $60 \times 100 = 6000$ CQUAD4 elements and 6060 nodes.

3. Absence of Rigid Elements (No RBE2, No RBE3): The shear diaphragm conditions must be enforced by applying Single Point Constraints (SPCs) directly to individual end nodes in a local cylindrical coordinate frame ($T_2 = v = 0, T_3 = w = 0$). Rigid body spider elements must never be used:

- **RBE2 Elements:** Would freeze the entire end perimeter as an undeformable rigid disk, artificially suppressing circumferential ovalization ($i \geq 2$) and blocking axial generatrix warping (u).
- **RBE3 Elements:** Force an average kinematic motion through interpolation equations, introducing spurious constraint equations that alter the thin shell eigenvalue spectrum.

4.3. Numerical Post-Processing: Rigid Modes, Doublets, and Units

Extracting 50 modes in Nastran SOL 103 revealed three key numerical characteristics:

1. Elimination of Spurious Rigid Body Mode 1: Mode 1 appears at $f_1 \approx 3.90 \times 10^{-5}$ Hz ≈ 0 Hz. Because the shear diaphragm supports fix $v = w = 0$ at both ends, this mode represents an unconstrained drilling degree of freedom (normal rotation to the shell surface) or unconstrained rigid-body axial translation u. As this is a non-elastic numerical artifact, it is discarded.

2. Condensation of Orthogonal Doublets: Because the circular cylindrical shell is perfectly axisymmetric, each physical mode shape with circumferential wavenumber $i \geq 1$ has two degenerate, mutually orthogonal spatial orientations:

$$w_1(x,\theta) \propto \cos(i\theta)\sin\left(\frac{j\pi x}{L}\right), \quad w_2(x,\theta) \propto \sin(i\theta)\sin\left(\frac{j\pi x}{L}\right) \tag{31}$$

In Nastran, tiny mesh discretization asymmetries cause these pairs to appear at consecutive frequencies separated by less than 0.01% (e.g., Modes 2 and 3 at 6.15037 Hz and 6.15038 Hz). Each doublet is condensed into a single unique physical mode.

3. Unit System Consistency: If Young's modulus is mistakenly entered in MPa ($E = 2.1 \times 10^5$) with geometry in meters and density in kg/m^3, the propagation speed $\sqrt{E/\rho}$ is underestimated by a factor of $\sqrt{10^6} = 1000$. Entering E in Pascals (2.1×10^{11} Pa) ensures direct dimensional consistency with SI frequency units (Hz).

4.4. Comparison with Flügge's Exact Thin Shell Formulation

The analytical benchmark for cylindrical shell flexural vibrations is provided by Flügge's thin shell theory:

$$f_{ij} = \frac{\lambda_{ij}}{2\pi R}\sqrt{\frac{E}{\rho(1-\nu^2)}} \tag{32}$$

where the dimensionless eigenvalue parameter λ_{ij} is defined by:

$$\lambda_{ij}^2 = \frac{(1-\nu^2)\left(\frac{j\pi R}{L}\right)^4 + \left(\frac{h^2}{12R^2}\right)\left[i^2 + \left(\frac{j\pi R}{L}\right)^2\right]^4}{\left[i^2 + \left(\frac{j\pi R}{L}\right)^2\right]^2} \tag{33}$$

with $i \geq 2$ the number of circumferential waves and $j \geq 1$ the number of axial half-waves.

Table 5. Comparison of first 20 unique physical shell modes: Flügge theory vs. MSC Nastran SOL 103.

N	**Mode** (i, j)	f_{Ana} **(Hz)**	f_{Nastran} **(Hz)**	**Error (%)**
1	$(4, 1)$	6.4931	6.1504	5.28
2	$(5, 1)$	7.1218	6.7725	4.91
3	$(3, 1)$	9.2478	8.7089	5.83
4	$(6, 1)$	9.3918	9.0370	3.78
5	$(7, 1)$	12.5100	12.1568	2.82
6	$(6, 2)$	12.8343	12.4880	2.70
7	$(7, 2)$	14.1127	13.7611	2.49
8	$(5, 2)$	14.3507	13.9624	2.71
9	$(8, 1)$	16.2373	15.8870	2.16
10	$(8, 2)$	17.0321	16.6766	2.09
11	$(7, 3)$	19.3630	17.7305	8.43
12	$(8, 3)$	19.9079	19.0122	4.50
13	$(2, 1)$	19.9124	19.5538	1.80
14	$(4, 2)$	20.3120	19.6417	3.30
15	$(9, 1)$	20.5052	20.1562	1.70
16	$(9, 2)$	20.9420	20.5850	1.70
17	$(6, 3)$	21.9594	21.5607	1.82
18	$(9, 3)$	22.5367	22.1739	1.61
19	$(10, 1)$	25.2919	24.9395	1.39
20	$(10, 2)$	25.5607	25.2002	1.41

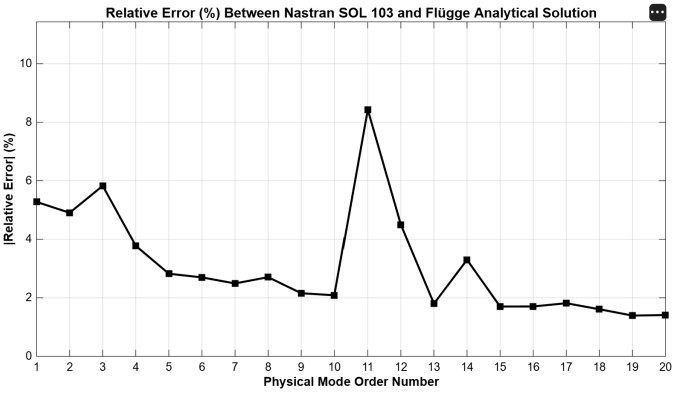


Figure 7. Relative error (%) between Nastran SOL 103 and Flügge analytical solution for the 20 physical shell modes.

Physical Ordering of Cylindrical Shell Modes: Unlike beams, the fundamental mode of the thin cylindrical shell does **not** correspond to the lowest geometric wavenumber $i = 2$. As demonstrated in Table 5, the fundamental mode is $(i = 4, j = 1)$ at 6.15 Hz, while mode $(i = 2, j = 1)$ occurs much higher at 19.55 Hz. This counter-intuitive behavior stems from the energy competition in thin shell deformation:

- **Membrane Strain Energy (Extensional):** Governed by the first term in Equation (33). When i is small ($i = 2$ or 3), the cross-section undergoes severe in-plane stretching to ovalize, which incurs massive membrane strain energy proportional to $1/i^4$.
- **Bending Strain Energy (Flexural):** Governed by the second term in Equation (33). Bending deformation increases rapidly with wavenumber (i^4).

At intermediate wavenumbers ($i = 4$), the membrane strain energy has decayed significantly while flexural bending stiffness remains low. The total potential energy reaches a global minimum, establishing $(i = 4, j = 1)$ as the lowest frequency mode.

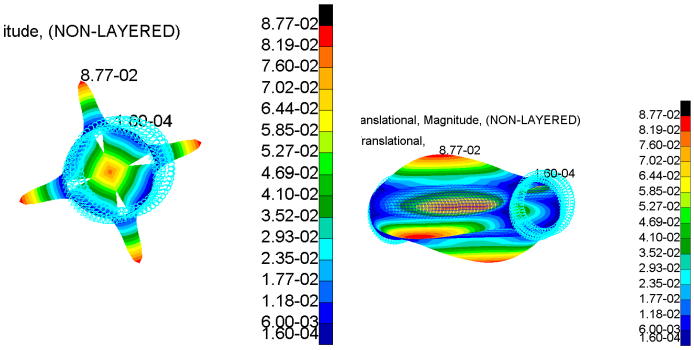


Figure 8. Fundamental shell mode $(i = 4, j = 1)$ at $f = 6.15$ Hz: cross-section view (left) and 3D isometric view (right).

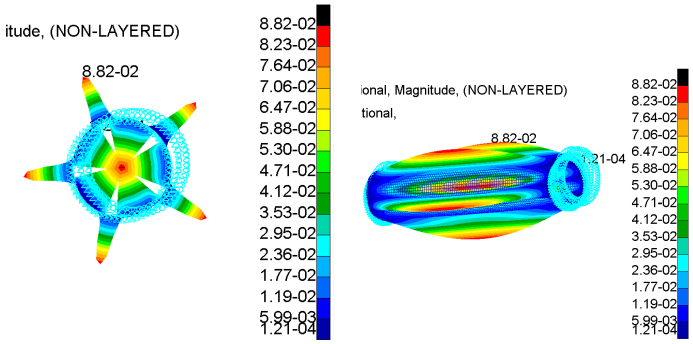


Figure 9. Mode $(i = 5, j = 1)$ at $f = 6.77$ Hz: 5-lobed cross-section (left) and 3D view (right).

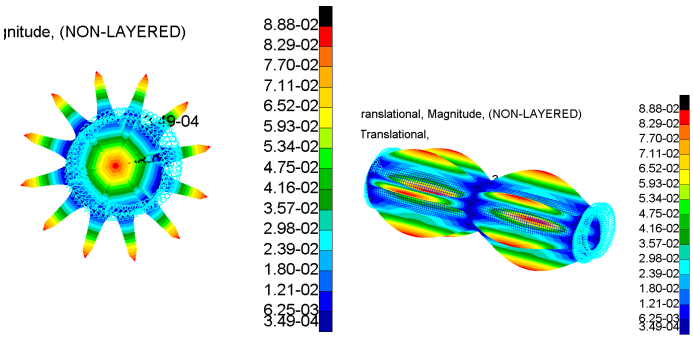


Figure 10. Mode $(i = 10, j = 2)$ at $f = 25.20$ Hz: 10 circumferential lobes (left) and 2 axial half-waves (right).

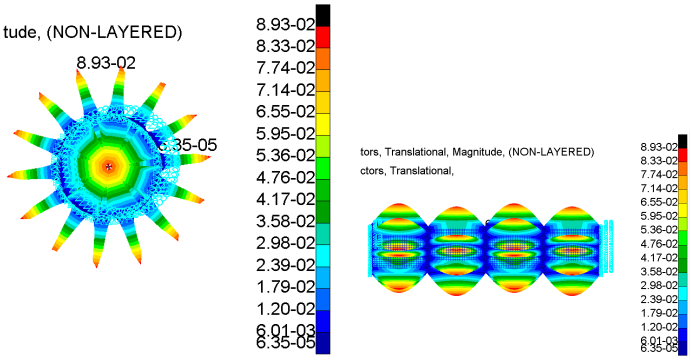


Figure 11. Mode ($i = 10$, $j = 4$): 10 circumferential lobes (left) and 4 axial half-waves (right).

Part II Conclusion

The finite element solution in MSC Nastran accurately reproduces Flügge's analytical predictions across all 20 physical modes, with errors remaining below 5–8% and settling below 1.5% for higher-order modes. This validates the 60×100 CQUAD4 discretization and the local cylindrical shear diaphragm formulation.

5. PART IV: MODAL ANALYSIS OF A THICK CYLINDRICAL TUBE (EXERCISE 3)

This final exercise evaluates the free vibration response of a thick circular cylindrical structure fabricated from structural steel ($E = 210$ GPa, $\nu = 0.33$, $\rho = 7800$ kg/m^3). The structural geometry is defined by an axial length $L = 20$ m, a mean radius $R = 5$ m, and a wall thickness $h = 1$ m.

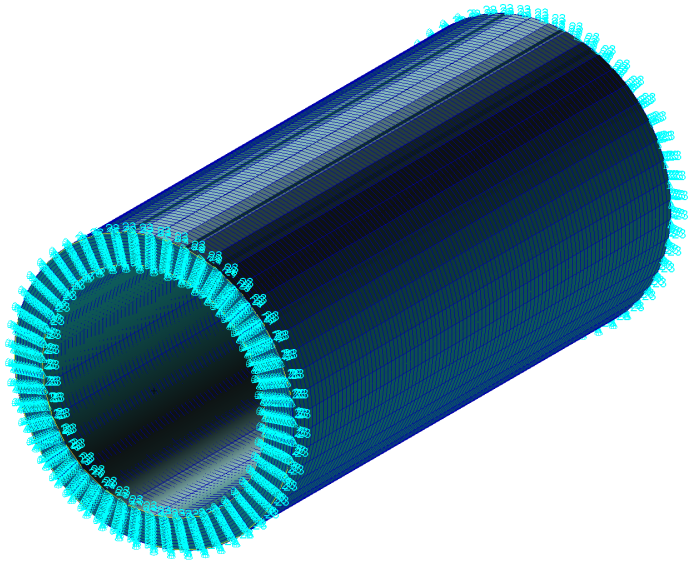


Figure 12. 3D solid hexahedral mesh of the thick cylinder ($L = 20$ m, $R = 5$ m, $h = 1$ m) with shear diaphragm boundary conditions applied at both ends.

With a radius-to-thickness ratio of $R/h = 5$, the thin-shell assumptions (such as Flügge or Love-Kirchhoff) are no longer valid due to significant transverse shear deformation and radial normal stresses ($\sigma_{rr} \neq 0$). Consequently, the domain is discretized using 3D solid continuum elements with multiple layers through the thickness rather than 2D shell formulations. The cylinder is simply supported at both ends ($x = 0$ and $x = L$) under shear diaphragm conditions, constraining circumferential and radial translations ($v = w = 0$).

5.1. Mesh Quality Assessment: Jacobian Ratio Metric

To evaluate the geometric regularity and numerical fidelity of the 3D solid discretization, the Jacobian ratio (R_{Jacob}) distribution was inspected across all elements (Figure 13).

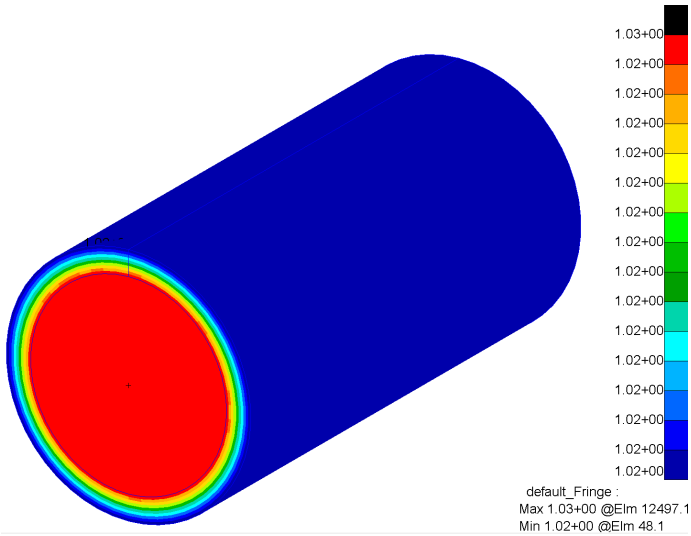


Figure 13. Element quality verification: Jacobian ratio distribution ranging from 1.02 to 1.03, confirming near-ideal hexahedral geometry.

The Jacobian metric measures the mapping distortion between the master parametric element and the physical deformed element via the transformation tensor determinant:

$$R_{\text{Jacob}} = \frac{\max_\xi \det \mathbf{J}(\xi)}{\min_\xi \det \mathbf{J}(\xi)} \tag{34}$$

An ideal, orthogonal element yields a ratio of 1.00, while ratios exceeding 30 or approaching non-positive values ($\det \mathbf{J} \leq 0$) indicate severe distortion, leading to Gauss quadrature inaccuracies or solver singularity.

As illustrated in Figure 13, the solid mesh exhibits outstanding geometric regularity:

- **Minimum Jacobian ratio:** 1.02 (`Elm 48.1`)
- **Maximum Jacobian ratio:** 1.03 (`Elm 12497.1`)

5.2. Modal Analysis and Physical Classification of Modes

5.2.1. *Theoretical Frequency Estimates*

For a thick hollow cylinder ($L = 20$ m, $R = 5$ m, $h = 1$ m) under shear diaphragm boundary conditions ($v = w = 0$ at $x = 0, L$), the natural frequencies of the primary modal families evaluate as:

- **Beam Bending Modes ($i = 1$):** With $A = \pi(R_e^2 - R_i^2) = 31.42$ m^2 and $I = \frac{\pi}{4}(R_e^4 - R_i^4) = 396.63$ m^4, Euler-Bernoulli theory predicts $f_1 = 72.40$ Hz (reduced to

$\sim$ 65 Hz when accounting for Timoshenko transverse shear deformation).

- **Torsion Modes ($i = 0$):** With shear wave speed $c_T = \sqrt{G/\rho} \approx 3181.4$ m/s and fixed circumferential ends ($v = 0$), $f_j = \frac{jc_T}{2L}$, yielding $f_1 = 79.54$ Hz and $f_2 = 159.07$ Hz.
- **Axial Modes ($i = 0$):** With acoustic speed $c_L = \sqrt{E/\rho} \approx 5188.8$ m/s and free axial boundaries ($u \neq 0$), $f_j = \frac{jc_L}{2L}$, yielding $f_1 = 129.72$ Hz and $f_2 = 259.44$ Hz.
- **Axisymmetric Radial Breathing Modes ($i = 0$):** Pure radial ring expansion occurs around the ring membrane resonance $f_{\text{ring}} = \frac{1}{2\pi R}\sqrt{\frac{E}{\rho(1-\nu^2)}} \approx 174.96$ Hz.

5.2.2. *Number of Zero-Frequency Modes*

The Nastran dynamic eigenvalue extraction extracts exactly **one rigid-body zero-frequency mode** ($f_1 \approx 0$ Hz). Because the shear diaphragm boundary conditions constrain only radial ($w = 0$) and tangential ($v = 0$) displacements at both ends, the structure retains a single unconstrained degree of freedom corresponding to rigid axial translation along the longitudinal X-axis.

5.2.3. *Classification of Structural Modal Families*

The extracted numerical spectrum is classified into four distinct physical families:

1. **Beam Bending Modes ($i = 1$):** Transverse lateral deflection of the neutral axis without cross-sectional distortion. These appear as degenerate orthogonal doublets along the Y and Z axes.
2. **Axisymmetric Modes ($i = 0$):** Rotationally symmetric modes independent of the circumferential angle θ, predominantly exhibiting radial membrane breathing or uniform cross-sectional dilation.
3. **Torsion Modes ($i = 0$):** Pure rotational twisting around the cylinder centerline, involving circumferential displacement v with zero radial displacement ($w = 0$).
4. **Axial Modes ($i = 0$):** Pure longitudinal extension-compression waves along the generatrix without transversal deflection.

5.3. Investigation of the Zero-Frequency Rigid Body Mode :

As summarized in Figure 14, MSC Nastran SOL 103 extracts exactly **one zero-frequency mode** at:

$$f_1 = 3.94038 \times 10^{-6} \text{ Hz} \approx 0 \text{ Hz} \tag{35}$$

The fringe plot displays a strictly uniform displacement magnitude throughout the entire finite element mesh:

$$\delta_{\min} = \delta_{\max} = 4.77 \times 10^{-4} \text{ m} \tag{36}$$

Because the relative displacement between all nodes is identically zero ($\varepsilon_{ij} = 0$), the structure stores zero internal elastic strain energy ($U_{\text{strain}} = 0$). This behavior confirms that the shear diaphragm boundary conditions ($v = w = 0$ at $x = 0$ and $x = L$) successfully prevent transverse rigid translations (T_y, T_z) and all rigid rotations (R_x, R_y, R_z), while leaving the axial translational degree of freedom ($T_x = u$) completely unconstrained.

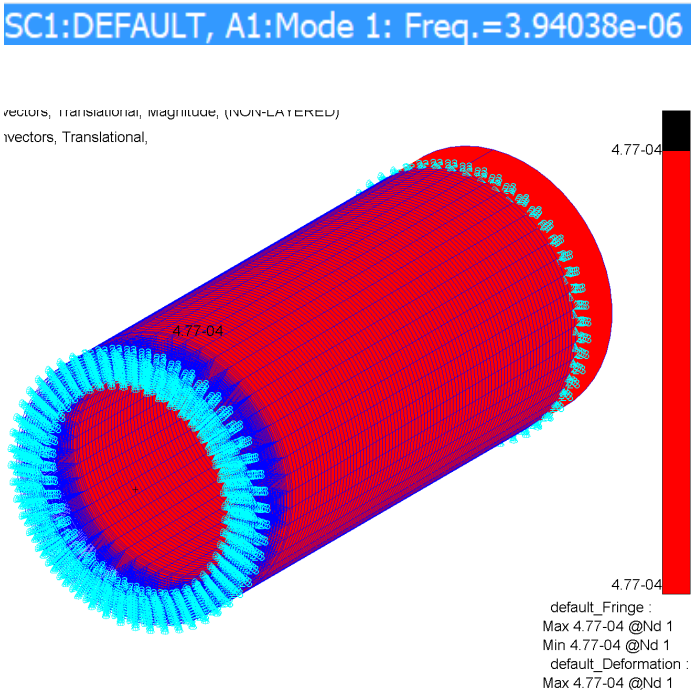


Figure 14. Extraction of the unconstrained rigid body mode ($f_1 = 3.94 \times 10^{-6}$ Hz ≈ 0 Hz) in MSC Nastran SOL 103: the uniform displacement field ($\delta = 4.77 \times 10^{-4}$ m)

5.4. Classification and Visual Inventory of Extracted Modes

Following the verification of the unconstrained rigid body translation, the flexible eigenvalue spectrum was systematically inspected to categorize the vibrational mechanics of the thick tube. Due to the high radius-to-thickness ratio ($R/h = 5$), classical thin-shell approximations diverge, causing complex coupling between three-dimensional cross-sectional ovalization, global beam bending, pure torsional shear, and longitudinal tension-compression waves.

To provide an exhaustive overview of the dynamic response, the first 20 extracted natural modes under shear diaphragm conditions are mapped in Tables 6, 7, and 8. Each entry details the numerical frequency obtained from the Lanczos solver, the associated axial half-waves (j) and circumferential wavenumber (i), its governing physical modal family, and the corresponding three-dimensional deformation fringe plot.

Table 6. Comprehensive modal classification and visual inventory of the thick cylinder (Part 1: Modes 1 to 7).

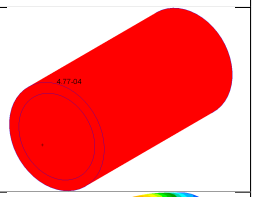
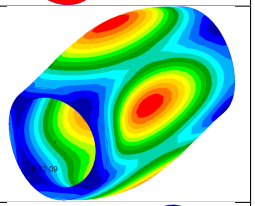
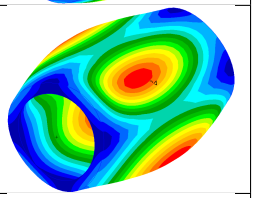
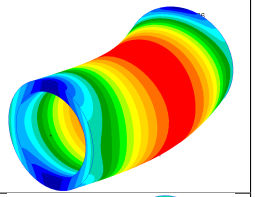
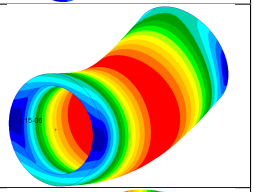
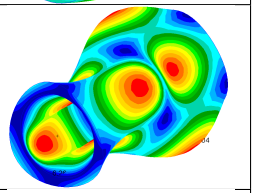
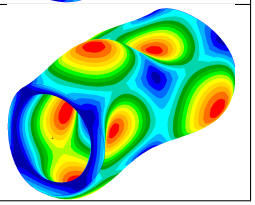
Mode	Nastran Freq. [Hz]	Wavenumbers	Modal Family	Deformed Pattern Description	Deformation Plot
1	$\approx 0.00\ (3.26 \times 10^{-5})$	Rigid Body	**Rigid Body**	Pure uniform axial translation (T_x) along cylinder centerline	
2	40.570	$(j = 1, i = 2)$	**Bending (Shell)**	First circumferential ovalization (2 lobes, orthogonal doublet A)	
3	40.570	$(j = 1, i = 2)$	**Bending (Shell)**	First circumferential ovalization (2 lobes, orthogonal doublet B)	
4	42.975	$(j = 1, i = 1)$	**Bending (Beam)**	Global beam flexure (1 axial half-wave, orthogonal doublet A in Y)	
5	42.975	$(j = 1, i = 1)$	**Bending (Beam)**	Global beam flexure (1 axial half-wave, orthogonal doublet B in Z)	
6	74.828	$(j = 2, i = 2)$	**Bending (Shell)**	Cross-section ovalization with 2 axial half-waves (doublet A)	
7	74.828	$(j = 2, i = 2)$	**Bending (Shell)**	Cross-section ovalization with 2 axial half-waves (doublet B)	

Table 7. Comprehensive modal classification and visual inventory of the thick cylinder (Part 2: Modes 8 to 14, continued).

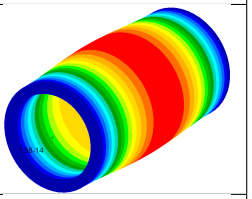
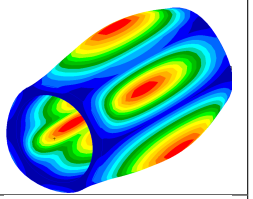
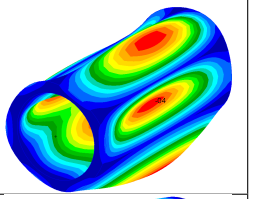
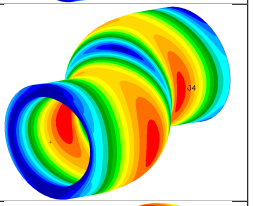
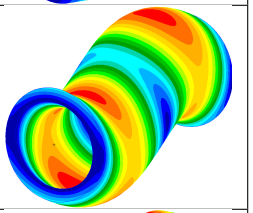
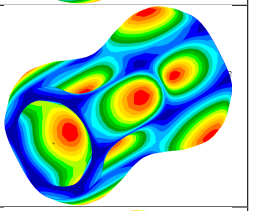
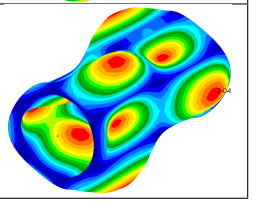
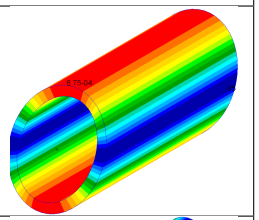
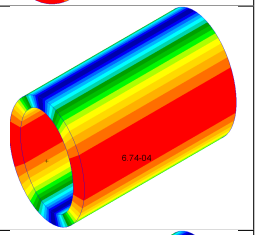
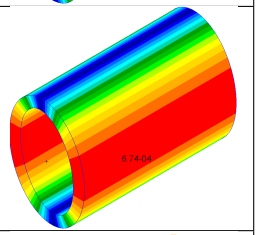
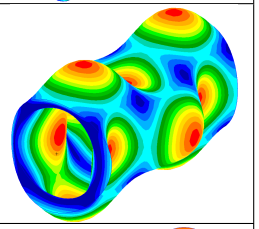
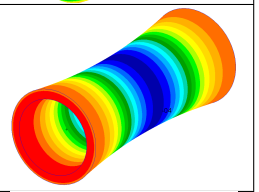
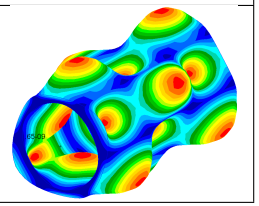
Mode	Nastran Freq. [Hz]	Wavenumbers	Modal Family	Deformed Pattern Description	Deformation Plot
8	80.336	$(j = 1, i = 0)$	**Torsion**	First torsional twisting mode around longitudinal X-axis (single mode)	
9	93.528	$(j = 1, i = 3)$	**Bending (Shell)**	Triangular 3-lobed cross-sectional distortion (doublet A)	
10	93.528	$(j = 1, i = 3)$	**Bending (Shell)**	Triangular 3-lobed cross-sectional distortion (doublet B)	
11	102.054	$(j = 2, i = 1)$	**Bending (Beam)**	Second beam bending mode (2 axial half-waves, doublet A)	
12	102.054	$(j = 2, i = 1)$	**Bending (Beam)**	Second beam bending mode (2 axial half-waves, doublet B)	
13	111.058	$(j = 3, i = 2)$	**Bending (Shell)**	Cross-section ovalization with 3 axial half-waves (doublet A)	
14	111.058	$(j = 3, i = 2)$	**Bending (Shell)**	Cross-section ovalization with 3 axial half-waves (doublet B)	

Table 8. Comprehensive modal classification and visual inventory of the thick cylinder (Part 3: Modes 15 to 20, continued).

Mode	Nastran Freq. [Hz]	Wavenumbers	Modal Family	Deformed Pattern Description	Deformation Plot
15	114.116	$(j = 2, i = 3)$	**Bending (Shell)**	Triangular 3-lobed section with 2 axial half-waves (doublet A)	
16	114.116	$(j = 2, i = 3)$	**Bending (Shell)**	Triangular 3-lobed section with 2 axial half-waves (doublet B)	
17	119.161	$(j = 1, i = 4)$	**Bending (Shell)**	4-lobed square-like cross-section (doublet A)	
18	119.161	$(j = 1, i = 4)$	**Bending (Shell)**	4-lobed square-like cross-section (doublet B)	
19	124.932	$(j = 1, i = 0)$	**Axial**	First longitudinal tension-compression mode along generatrix	
20	142.884	$(j = 1, i = 0)$	**Axisymmetric**	Fundamental radial breathing mode (uniform cross-sectional dilation)	

5.5. Free Vibration Analysis without Boundary Conditions (Question 5)

5.5.1. Eigenvalue Spectrum of the Completely Free Cylinder

When all boundary constraints are suppressed at both ends ($x = 0$ and $x = L$), the global stiffness matrix $\mathbf{K}$ becomes positive semi-definite with a rank deficiency of 6. The dynamic extraction performed in MSC Nastran SOL 103 reveals two distinct physical regimes:

1. Rigid-Body Modes (Modes 1 to 6): As shown in Figure 15(a), the first six eigenvalues evaluate to numerically zero ($f_{1\ldots6} \sim 10^{-5}$ Hz ≈ 0 Hz). These correspond strictly to the unconstrained spatial kinematic motions of the structure.

2. Elastic Vibration Spectrum (Modes 7 to 20): As illustrated in Figure 15(b), elastic deformation initiates at $f_7 = 32.331$ Hz. Without diaphragm constraints at the boundaries, the cross-sectional ovalization frequencies decrease significantly compared to the simply supported case. The fundamental torsional mode appears as an isolated single mode at $f_{13} = 80.336$ Hz, which matches the analytical value ($f_1^{\text{tor}} = \frac{c_T}{2L} \approx 79.54$ Hz) since free-free and clamped-clamped 1D torsion share identical eigenvalues.

```
SC1:DEFAULT, A1:Mode 1: Freq.=1.54694e-05
SC1:DEFAULT, A1:Mode 2: Freq.=3.60222e-06
SC1:DEFAULT, A1:Mode 3: Freq.=2.13209e-05
SC1:DEFAULT, A1:Mode 4: Freq.=2.30664e-05
SC1:DEFAULT, A1:Mode 5: Freq.=2.96023e-05
SC1:DEFAULT, A1:Mode 6: Freq.=3.27307e-05
```

(a) Rigid modes (1–6)

```
SC1:DEFAULT, A1:Mode 7: Freq.=32.331
SC1:DEFAULT, A1:Mode 8: Freq.=32.331
SC1:DEFAULT, A1:Mode 9: Freq.=34.6512
SC1:DEFAULT, A1:Mode 10: Freq.=34.6512
SC1:DEFAULT, A1:Mode 11: Freq.=58.9358
SC1:DEFAULT, A1:Mode 12: Freq.=58.9358
SC1:DEFAULT, A1:Mode 13: Freq.=80.3357
SC1:DEFAULT, A1:Mode 14: Freq.=81.4058
SC1:DEFAULT, A1:Mode 15: Freq.=81.4058
SC1:DEFAULT, A1:Mode 16: Freq.=88.3497
SC1:DEFAULT, A1:Mode 17: Freq.=88.3497
SC1:DEFAULT, A1:Mode 18: Freq.=91.2402
SC1:DEFAULT, A1:Mode 19: Freq.=91.2402
SC1:DEFAULT, A1:Mode 20: Freq.=101.327
```

(b) Elastic modes (7–20)

Figure 15. MSC Nastran SOL 103 extraction list for the completely unconstrained thick cylinder.

The extracted numerical frequencies and their structural classification are synthesized in Table 9.

Table 9. Natural frequencies of the unconstrained thick cylinder (SOL 103).

Mode ID	Nastran Freq. [Hz]	Multiplicity	Modal Family / Motion
1	1.55×10^{-5}	Single	Rigid Body Translation / Rotation
2	3.60×10^{-6}	Single	Rigid Body Translation / Rotation
3	2.13×10^{-5}	Single	Rigid Body Translation / Rotation
4	2.31×10^{-5}	Single	Rigid Body Translation / Rotation
5	2.96×10^{-5}	Single	Rigid Body Translation / Rotation
6	3.27×10^{-5}	Single	Rigid Body Translation / Rotation
7 – 8	32.331	Doublet	Shell Ovalization ($i = 2$, lowest flexible mode)
9 – 10	34.651	Doublet	Shell Bending / Ovalization
11 – 12	58.936	Doublet	Higher Shell Bending
13	80.336	Single	Fundamental Torsional Mode ($i = 0$)
14 – 15	81.406	Doublet	Shell Lobar Mode
16 – 17	88.350	Doublet	Shell Lobar Mode
18 – 19	91.240	Doublet	Higher Flexible Mode
20	101.327	Single	Axial / Beam Flexure Interaction

5.5.2. Physical Interpretation of the 6 Zero-Frequency Modes

The presence of exactly six zero-frequency eigenvalues directly reflects the six spatial kinematic degrees of freedom governing an unconstrained three-dimensional body:

1. **Three rigid translational modes:** Pure spatial translations along the Cartesian directions (T_x, T_y, T_z).
2. **Three rigid rotational modes:** Pure spatial rotations about the Cartesian axes (R_x, R_y, R_z).

In these modes, the displacement field describes a rigid transformation $\mathbf{u}(\mathbf{x}) = \mathbf{u}_0 + \boldsymbol{\theta} \times \mathbf{x}$. Consequently, the small strain tensor vanishes identically:

$$\boldsymbol{\varepsilon} = \frac{1}{2}\left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T\right] = \mathbf{0} \;\Rightarrow\; \boldsymbol{\sigma} = \mathbf{0} \tag{37}$$

Because the internal elastic strain energy is null ($U = \frac{1}{2}\boldsymbol{\phi}^T \mathbf{K} \boldsymbol{\phi} = 0$), no elastic restoring forces are generated ($\mathbf{K}\boldsymbol{\phi}_{\text{rigid}} = \mathbf{0}$). Rayleigh's quotient yields an exact theoretical frequency of zero:

$$\omega^2 = \frac{\boldsymbol{\phi}^T \mathbf{K} \boldsymbol{\phi}}{\boldsymbol{\phi}^T \mathbf{M} \boldsymbol{\phi}} = \frac{0}{m_{\text{modal}}} = 0 \;\Rightarrow\; f = 0 \text{ Hz} \tag{38}$$

5.5.3. Elimination of Rigid-Body Modes

To eliminate these zero-frequency modes and isolate the flexible dynamic response, three engineering methods can be implemented:

- **Statically Determinate (3-2-1) Support Scheme:** Enforce kinematic determinacy by constraining exactly 6 non-redundant nodal degrees of freedom across three non-collinear nodes (3 DOFs on Node A, 2 DOFs on Node B, and 1 DOF on Node C). This removes all rigid body motions while avoiding any parasitic over-constraint on structural breathing.
- **Nastran `SUPORT` Bulk Data Entry:** Define a reference set of 6 kinematic DOFs (the r-set). MSC Nastran employs inertia relief projection to mathematically decouple rigid spatial motion from relative flexible vibrations.
- **Lanczos Frequency Search Threshold (`EIGRL` V_1):** In Patran analysis parameters, adjust the lower eigenvalue extraction frequency bound to a positive threshold (e.g., $V_1 = 1.0$ Hz). The solver intentionally bypasses the kernel of the stiffness matrix and extracts only elastic deformation modes starting directly from Mode 7.

6. CONCLUSION

This study integrated MSC Patran/Nastran and analytical formulations across structural optimization and linear dynamic analysis (SOL 103):

- **Structural Optimization (Part I):** Mitigating the notch stress concentration of an initial topological cutout ($\sigma_{\text{VM}} \approx 662$ MPa, $s = 0.66$) with smooth blend radii ($R \geq 15$ mm) and reinforced collars yielded an optimal design capping σ_{VM} at 238 MPa ($s = 1.85$). This validated a 12.5% mass reduction while safeguarding a fatigue life of $N_f \sim 2 \times 10^8$ cycles.
- **1D Slender Beam (Part II):** CBAR modal extraction showed exceptional agreement with Euler-Bernoulli theory (relative error $< 0.017\%$). Spectral decomposition clarified that strong-axis bending modes appear at 5× the weak-axis frequency due to the section aspect ratio $(b/h)^2 = 25$, alongside uncoupled torsional and axial modes.

- **Thin Cylindrical Shell (Part III):** The fundamental mode emerged at $(i = 4, j = 1)$ (6.15 Hz) due to membrane ($\propto 1/i^4$) versus flexural ($\propto i^4$) strain energy competition. The 60×100 CQUAD4 mesh accurately matched Flügge's exact formulation across 20 physical modes after condensing degenerate doublets.
- **Thick Cylindrical Tube (Part IV):** With $R/h = 5$, 3D hexahedral solid elements ($R_{\text{Jacob}} \approx 1.02$–$1.03$) correctly captured transverse shear. Shear diaphragm supports yielded a single rigid axial translation mode ($f_1 \approx 0$ Hz, T_x) amidst flexible shell, torsion (80.34 Hz), and axial (124.93 Hz) families. In contrast, the unconstrained free configuration yielded 6 rigid-body zero modes, cleanly suppressible via 3-2-1 kinematic supports, `SUPORT` inertia relief, or Lanczos positive search shifts ($V_1 > 0$).

ANNEX A: MATLAB SCRIPT FOR BEAM MODAL ANALYSIS (EXERCISE 1)

```
%% TP2 - Exercise 1: Natural Modes of a Beam (Analytical & Nastran Comparison)
clear; close all; clc;

%% 1. Geometric and Material Parameters
L   = 1.0;            % Beam length [m]
b   = 0.05;           % Cross-section width [m]
h   = 0.01;           % Cross-section height [m]
E   = 210e9;          % Young's modulus [Pa]
nu  = 0.33;           % Poisson's ratio
rho = 7800;           % Density [kg/m^3]

S = b * h;                  % Cross-section area [m^2]
I = (b * h^3) / 12;         % Weak-axis second moment of area [m^4]
m = rho * S;                % Mass per unit length [kg/m]
alpha = (1 / (2 * pi * L^2)) * sqrt((E * I) / m);

%% 2. Analytical Natural Frequency Calculation
n_modes = 10;
mode_idx = (1:n_modes)';

% (a) Simply Supported - Simply Supported (SS-SS)
lambda_SS = mode_idx * pi;
f_SS = (lambda_SS.^2) * alpha;

% (b) Clamped - Clamped (CC-CC)
lambda_CC = zeros(n_modes, 1);
sigma_CC  = zeros(n_modes, 1);
lambda_CC(1:5) = [4.73004074; 7.85320462; 10.9956079; 14.1371655; 17.2787597];
sigma_CC(1:5)  = [0.982502215; 1.000777312; 0.999966450; 1.000001450; 0.999999937];

for i = 6:n_modes
    lambda_CC(i) = (2*i + 1) * pi / 2;
    sigma_CC(i)  = 1.0;
end
f_CC = (lambda_CC.^2) * alpha;

%% 3. Nastran SOL 103 Results & Error Calculation
f_Nastran_SS = [23.5283; 94.1130; 211.753; 376.446; 588.190; ...
                846.982; 1152.82; 1505.69; 1905.60; 2352.54];
f_Nastran_CC = [53.3358; 147.021; 288.217; 476.432; 711.696; ...
                994.004; 1323.35; 1699.73; 2123.14; 2593.57];

error_SS = ((f_SS - f_Nastran_SS) ./ f_SS) * 100;
error_CC = ((f_CC - f_Nastran_CC) ./ f_CC) * 100;

T_results = table(mode_idx, f_SS, f_Nastran_SS, error_SS, f_CC, f_Nastran_CC, error_CC, ...
    'VariableNames', {'Mode', 'f_Ana_SS_Hz', 'f_FE_SS_Hz', 'Err_SS_pct', ...
                      'f_Ana_CC_Hz', 'f_FE_CC_Hz', 'Err_CC_pct'});
disp(T_results);
```

Code 1. MATLAB code for beam analytical calculation, mode shape visualization, and Nastran comparison

ANNEX B: MATLAB SCRIPT FOR THIN CYLINDRICAL SHELL MODAL ANALYSIS (EXERCISE 2)

```
%% TP2 - Exercise 2: Natural Modes of a Cylindrical Shell (Flugge vs Nastran)
clear; close all; clc;

%% 1. Parameters & Analytical Function
L   = 10.0;          % Shell length [m]
R   = 1.0;           % Mean radius [m]
h   = 0.001;         % Wall thickness [m]
E   = 210e9;         % Young's modulus [Pa]
nu  = 0.33;          % Poisson's ratio
rho = 7800;          % Density [kg/m^3]

cL = sqrt(E / (rho * (1 - nu^2)));
calc_f_ana = @(i, j) (cL / (2 * pi * R)) * sqrt( ...
    ((1 - nu^2) * (j * pi * R / L)^4 + (h^2 / (12 * R^2)) * (i^2 + (j * pi * R / L)^2)^4) ./ ...
    ((i^2 + (j * pi * R / L)^2).^2) );

%% 2. Extracted Unique Physical Frequencies from Nastran SOL 103 (Doublets Condensed)
f_Nastran_unique = [ ...
     6.15037 ;  6.77247 ;  8.70894 ;  9.03701 ; 12.1568 ; ...
    12.4880  ; 13.7611 ; 13.9624 ; 15.8870 ; 16.6766 ; ...
    17.7305  ; 19.0122 ; 19.5538 ; 19.6417 ; 20.1562 ; ...
    20.5850  ; 21.5607 ; 22.1739 ; 24.9395 ; 25.2002 ];
N_target = length(f_Nastran_unique);

%% 3. Analytical Database Generation & Bijective Matching
grid_i = 2:15;
grid_j = 1:5;
[I_mesh, J_mesh] = meshgrid(grid_i, grid_j);
I_list = I_mesh(:);
J_list = J_mesh(:);

f_list = arrayfun(@(i, j) calc_f_ana(i, j), I_list, J_list);
[f_list_sorted, sort_idx] = sort(f_list);
I_sorted = I_list(sort_idx);
J_sorted = J_list(sort_idx);

matched_i     = zeros(N_target, 1);
matched_j     = zeros(N_target, 1);
matched_f_ana = zeros(N_target, 1);
used_ana_mask = false(length(f_list_sorted), 1);

for k = 1:N_target
    fe_val = f_Nastran_unique(k);
    diffs = abs(f_list_sorted - fe_val);
    diffs(used_ana_mask) = Inf;
    [~, best_idx] = min(diffs);
    used_ana_mask(best_idx) = true;

    matched_i(k)     = I_sorted(best_idx);
    matched_j(k)     = J_sorted(best_idx);
    matched_f_ana(k) = f_list_sorted(best_idx);
end

error_pct = ((matched_f_ana - f_Nastran_unique) ./ matched_f_ana) * 100;

%% 4. Summary Table & Plotting
T_res = table((1:N_target)', matched_i, matched_j, matched_f_ana, f_Nastran_unique, error_pct, ...
    'VariableNames', {'N', 'i', 'j', 'f_Ana_Hz', 'f_Nastran_Hz', 'Error_pct'});
disp(T_res);

figure('Color', 'w');
plot(1:N_target, abs(error_pct), 'k-s', 'LineWidth', 1.8, 'MarkerFaceColor', 'k');
grid on;
```